Setting the LS machine's patch size on the RHM

The length scale is the whole story -- and it must be chosen per site, not per step

The criterion "largest window matching any training patch" beats the best FIXED window in 12/12 cells, median 12x.

Every floor on top of it loses: the entropy floor h_min in 10/10 cells, and its count rival n_min at both P rungs. Rejection only shortens.

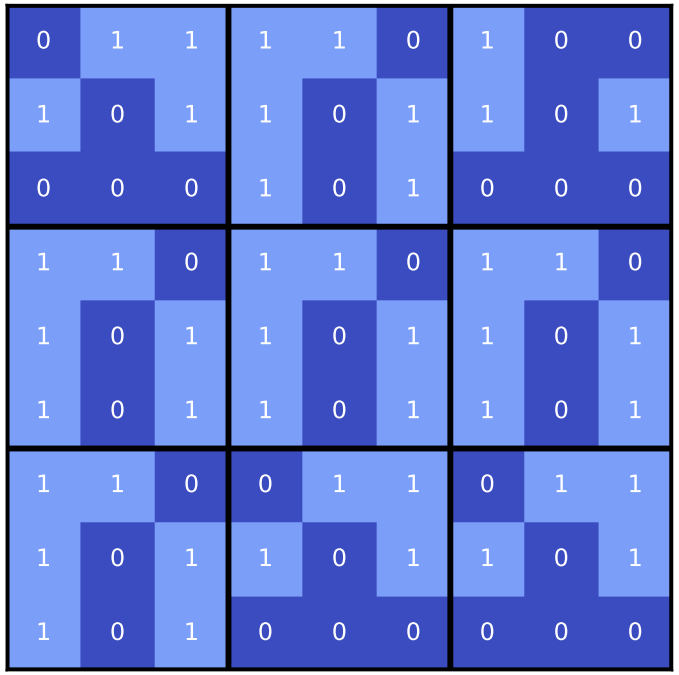
Replacing the per-site choice by its own induced k(t) schedule, at the SAME mean k, keeps only 15% of the success (9/9 cells).

Position explains none of the choice ($R^2 < 0.001$); local reveal count explains at most 0.47. The rest is irreducibly content-dependent.

s=3, L=2 is the first board where this question is even askable: at s=2 the only rung between U1 and the grid side is U1 itself.

Tokenization: what a "sample" is, and what a window sees

one sample: 9x9 grid of tokens in {0..1}
black lines bound the 3x3 level-1 blocks



Tokens. Every site holds one of v = 2 symbols. There is no intensity and no ordering: token 2 is not "between" 1 and 3. One grid is 9x9 = 81 categorical sites.

Grammar. A root symbol expands L = 2 times; each symbol is replaced by one of m synonymous 3x3 "words", drawn once and then fixed. A grid is grammatical iff some root derives it. |R_L| = v*m^(1+s^2), so m^10 at s=3.

Diffusion. Absorbing-state ("erasure"): each site is independently replaced by a MASK token with probability p(t). Unmasked sites are never corrupted, so exact Bayes denoising IS patch matching -- no score approximation enters, and any gap from optimal is the window choice, not a fitting error.

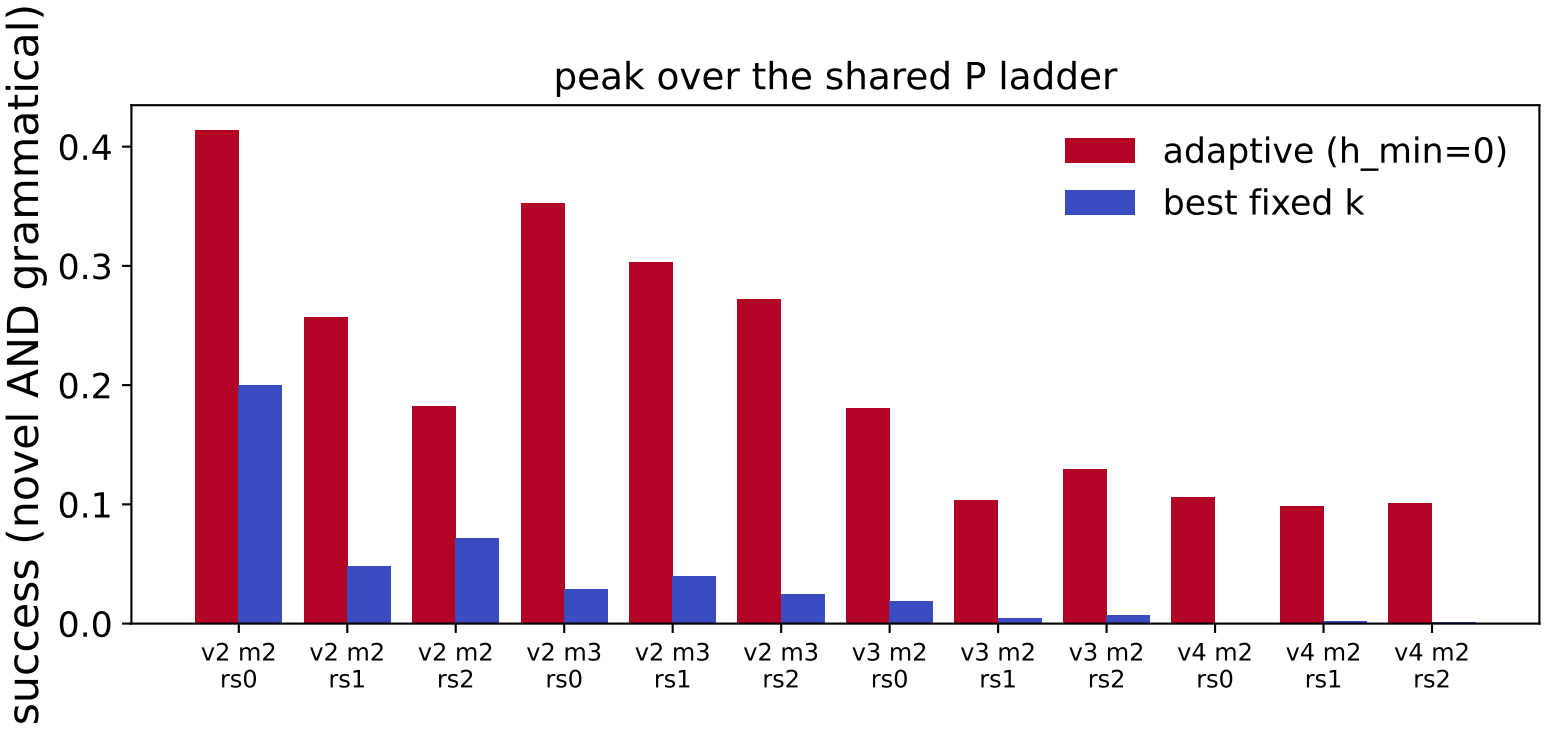
LS machine. The posterior at a masked site is the empirical distribution of the centre tokens of all training kxk patches agreeing with the observed sites in that window (Laplace alpha=1e-3). Sites are revealed one at a time.

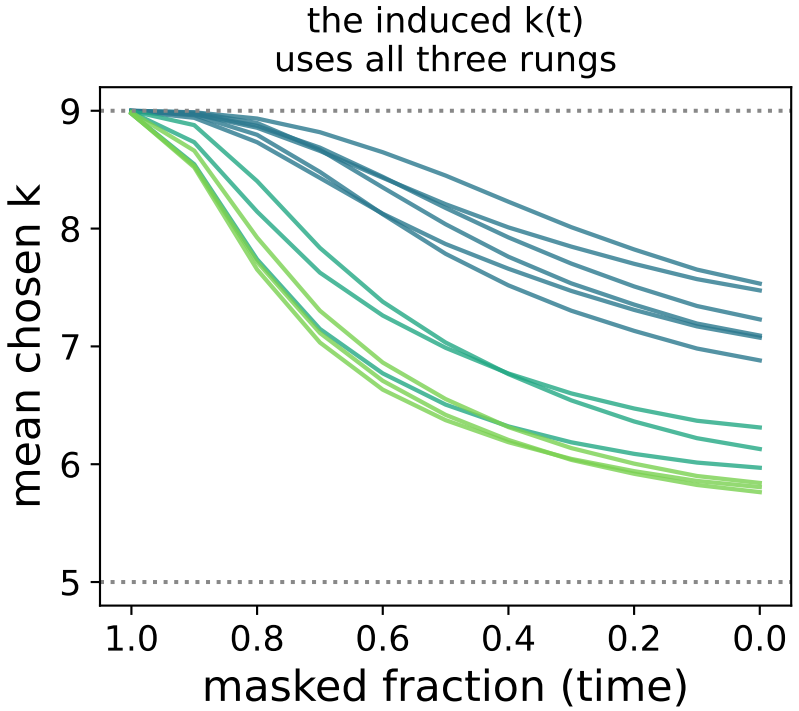
Why s=3. U1 = 2s^(L-1)-1 is the smallest window that can contain a whole top-level child block. At s=2, U1 = side-1, so the only odd rung in [U1, side] is U1 itself and there is no schedule to choose. At s=3, L=2 the board is 9x9 with three real rungs: 5, 7, 9.

Success (hshunt's criterion): a grid counts only if it is NOT in the training set AND parses completely.

success = grammatical AND novel

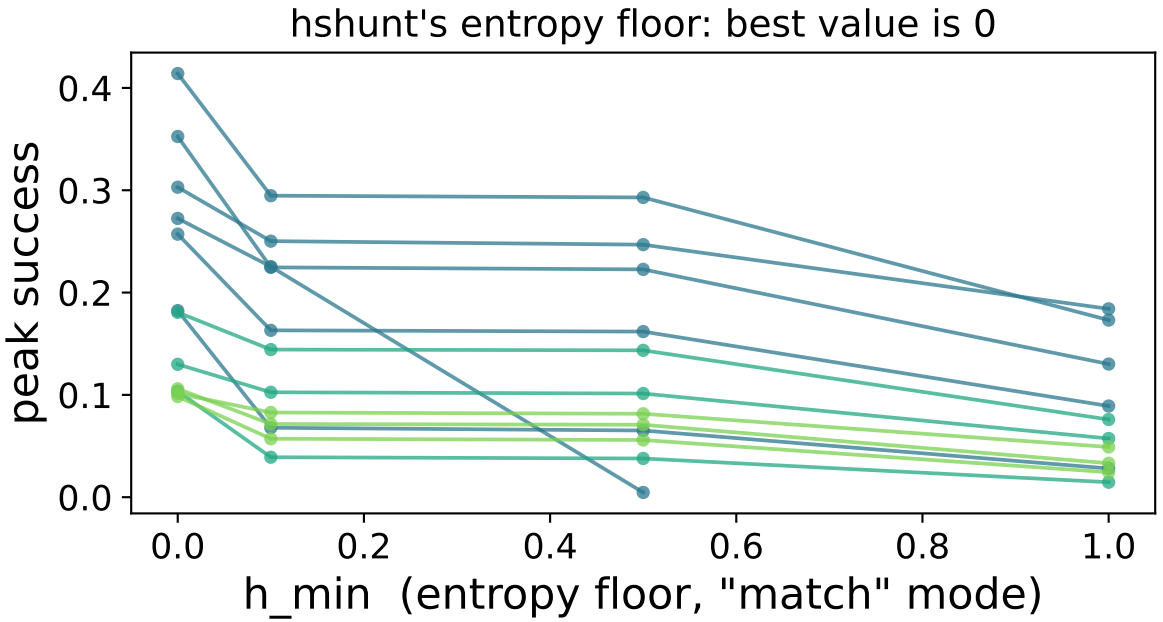
1. The window choice matters: adaptive beats every fixed k

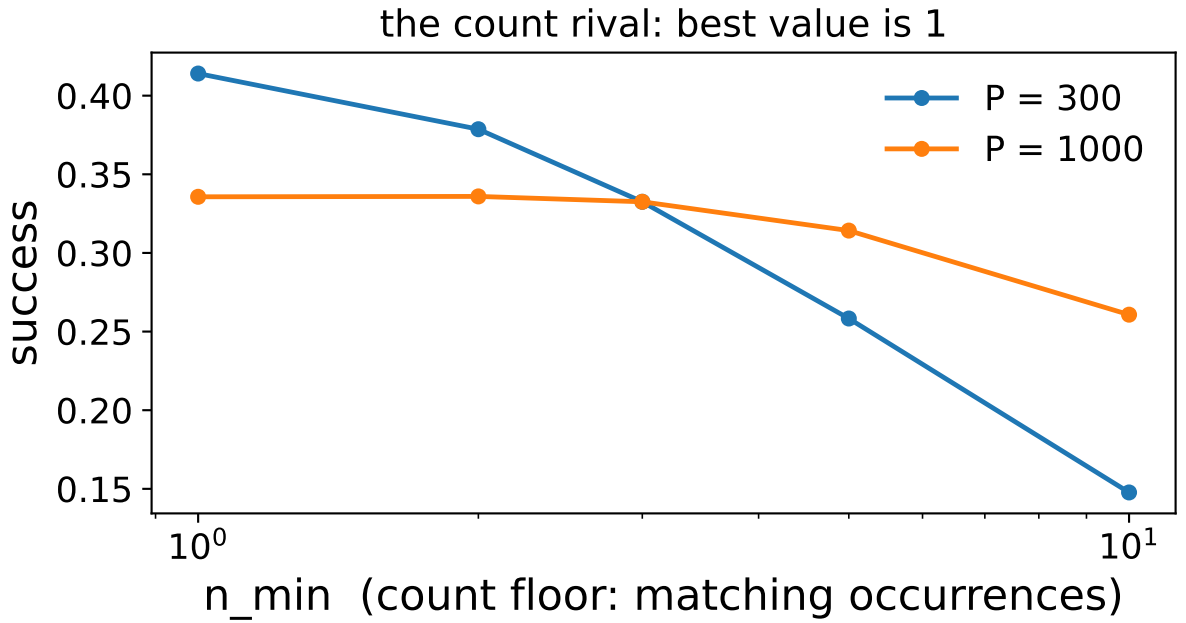





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Adaptive beats the best fixed k in 12 of 12 cells; median ratio 12x.
The fixed optimum sits at U1 = 5 in only 2 of 12 -- it prefers k = 7, so the
machine wants a window LARGER than the smallest one that can hold a child block.
k(t) falls from 9 as the grid fills, and its span grows with the vocabulary v
(about 1.5 at v=2, 2.9 at v=3, 3.2 at v=4): a richer grammar needs to back off
further as the grid fills.
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2. Neither floor earns its keep -- the bare support test wins



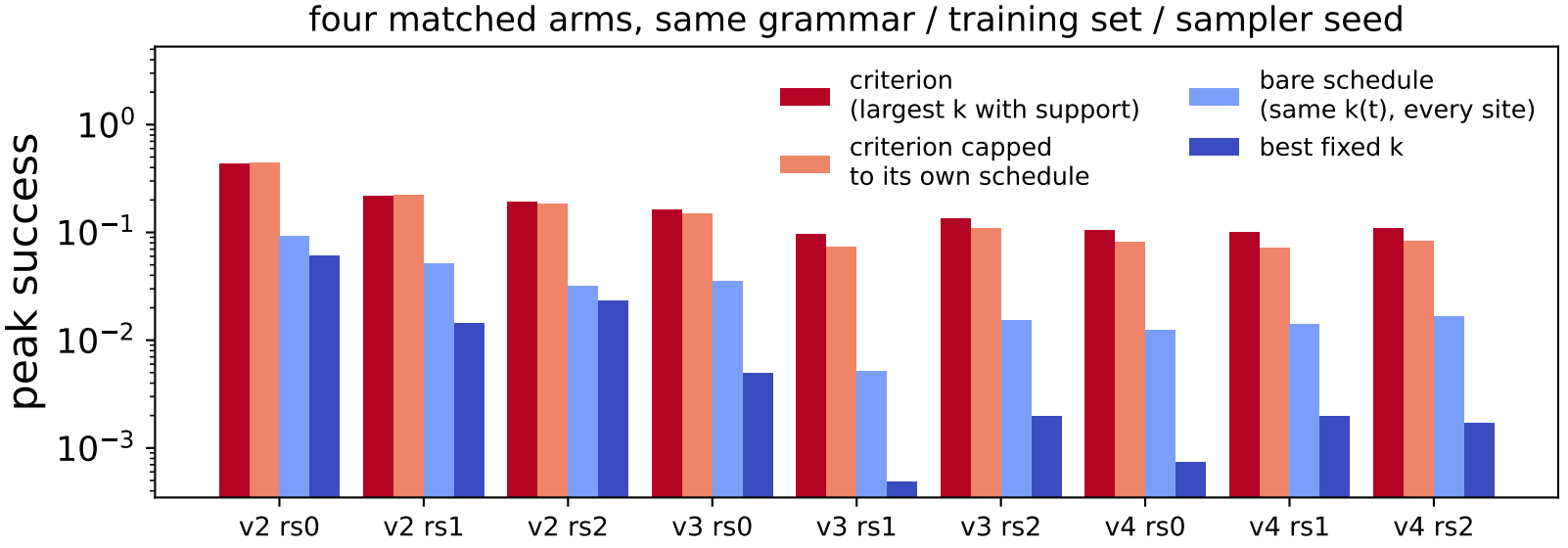


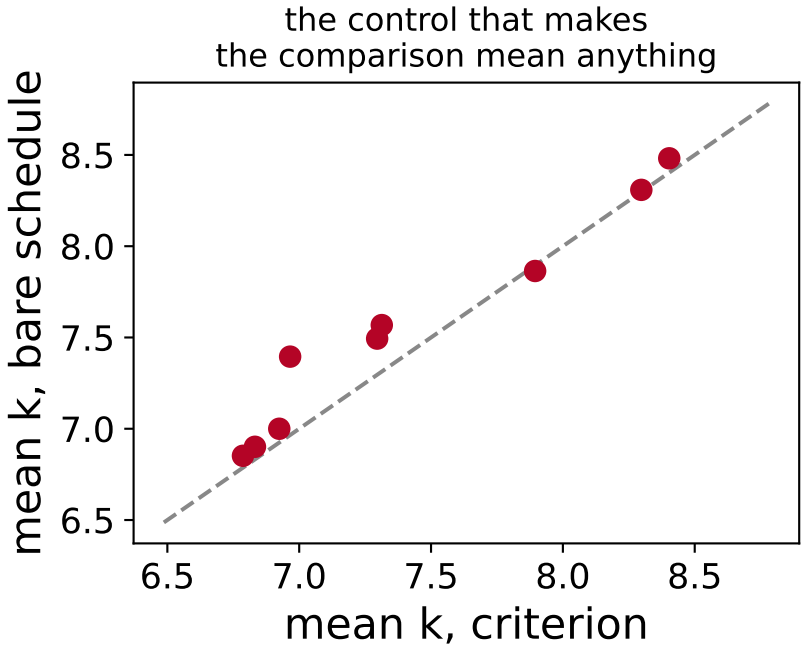

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The machine accepts a window when  n_match > 0  AND  h >= h_min.
In "match" mode h = log(effective number of matching training patches), so
h_min is a support-DIVERSITY floor and h_min = 0 is bare support-EXISTENCE.

n_min is the genuine rival, not a reparametrisation: n_match counts matching
training OCCURRENCES, h_match measures their DIVERSITY. A window matched by
fifty copies of one patch has n_match = 50 but h_match = 0 -- the entropy floor
throws it away, a count floor keeps it. Both lose anyway.

One mechanism explains both: rejecting a window can only send the site to a
SHORTER one, and shorter is less grammatical. The floors buy novelty and sell
grammaticality at a bad rate (n_min 1 -> 10: novel +0.14, grammatical -0.40).
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3. Nor is it a schedule: the SAME k(t), applied per site, is worth 4.7x



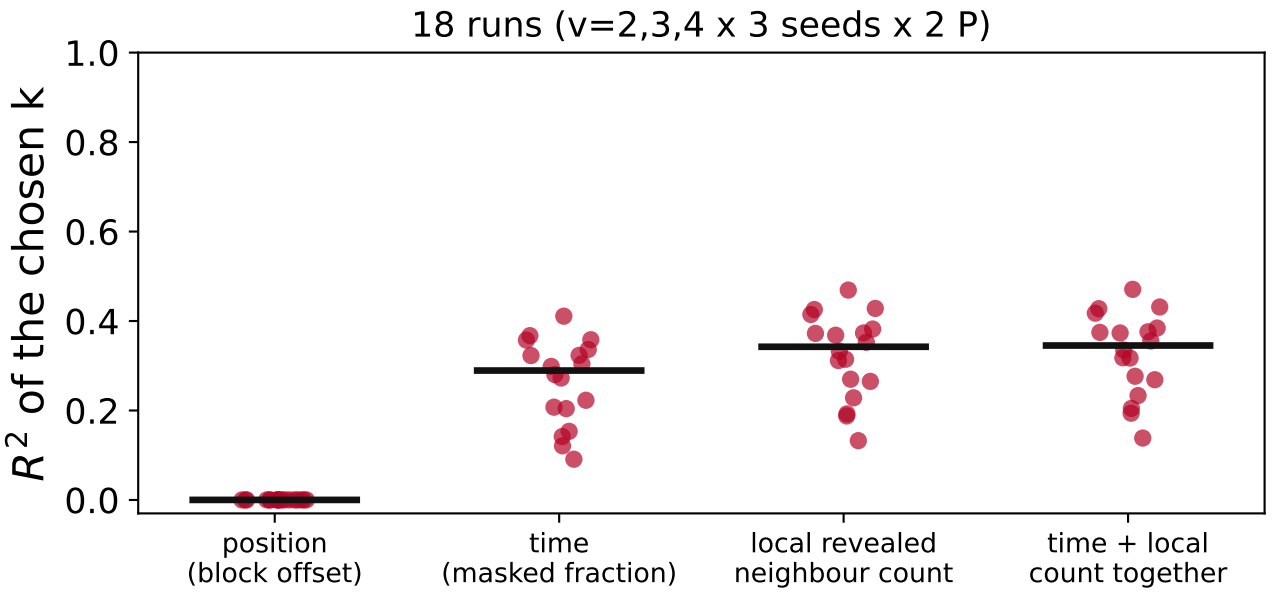


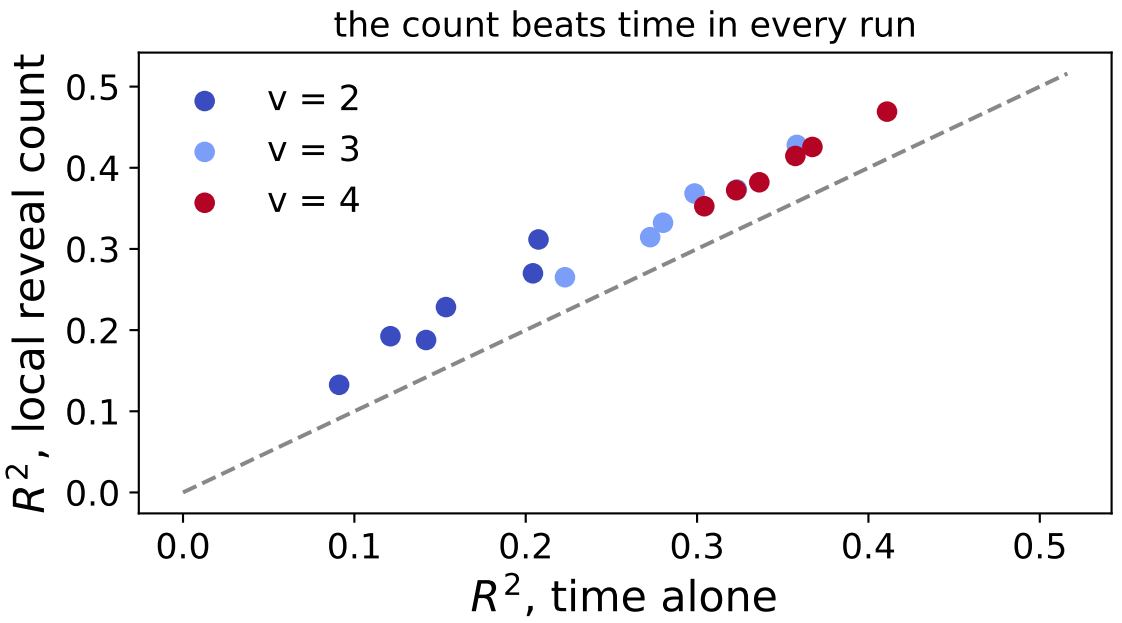
The control schedule is derived from the adaptive arm's OWN recorded trace -- round its mean k(t) to the available rungs -- so it cannot be rigged by picking a schedule that was always going to lose.

Adaptive beats the bare schedule in 9/9 cells and the best fixed k in 9/9. The schedule recovers a median 15% of adaptive success WHILE SPENDING THE SAME WINDOW BUDGET: median gap in mean k is 0.08 (right panel, on the diagonal).

So the win is not which window sizes get used on average. It is which SITE gets which size. Capping the criterion to its own schedule is free at v=2 but costs 8-30% at v=3,4, so the per-site freedom matters more as the grammar grows.

4. What the per-site choice actually depends on



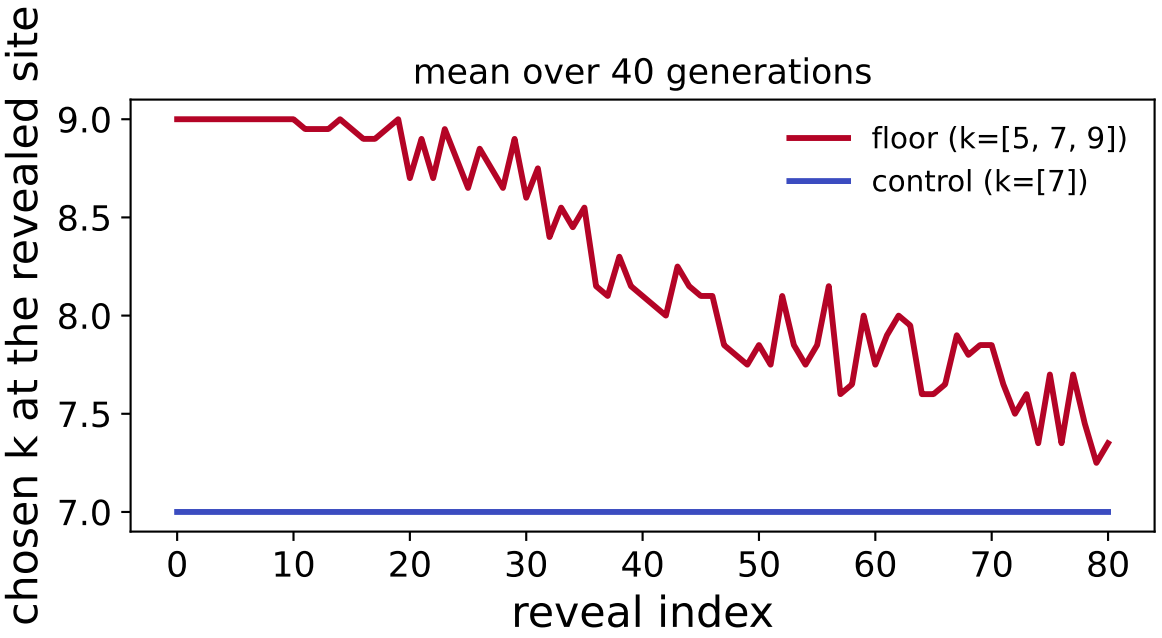


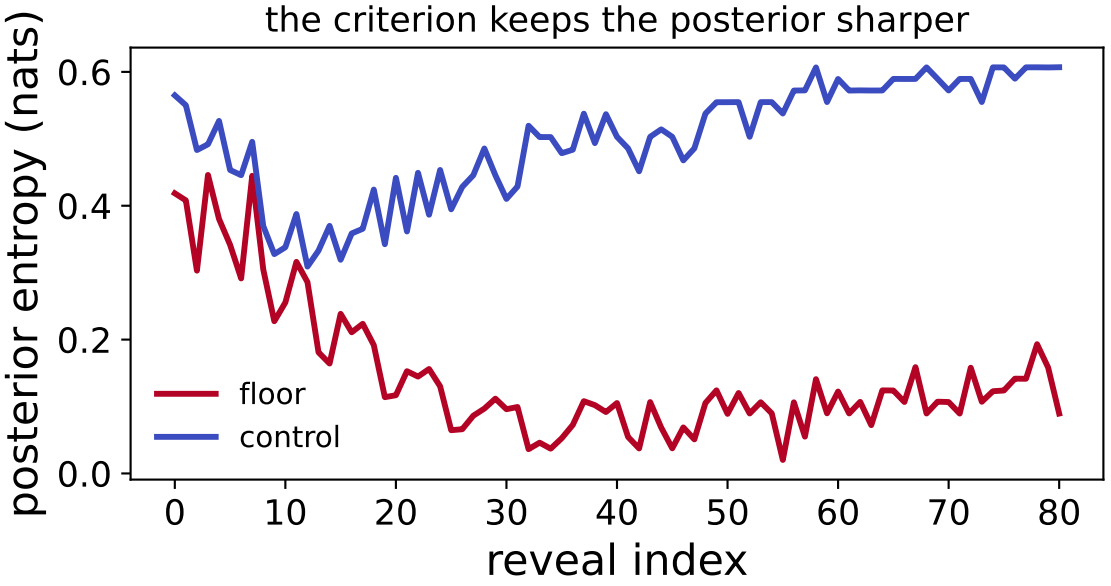
Position explains nothing. Regressing the chosen k on the site's offset within its level-1 block gives R^2 < 0.001 in all 18 runs, and adds nothing jointly with time. So no lookup table in (position, time) can replace the criterion -- and the window choice carries no signature of how it straddles the grammar blocks.

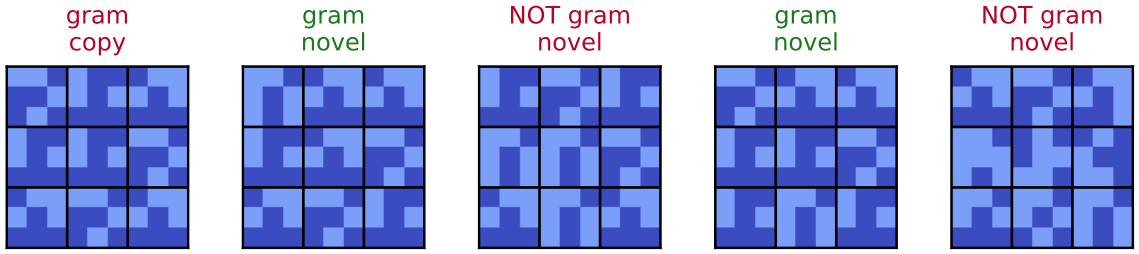
The local revealed-neighbour count (how many of the 9x9 around a site are fixed) beats time in 18/18 runs and SUBSUMES it: adding time on top gains only .005-.013. The k(t) schedule is therefore a consequence of reveal density, not a separate effect -- which is why page 3's bare schedule fails at matched mean k.

But it caps at 0.47. The remaining >= 53% turns on WHICH tokens were revealed, not how many. The criterion is irreducibly content-dependent.

5. Example traces: the window size moving along one generation







criterion output, the FIRST 5 generations (not selected)
v2 m2 rs0, P=300. Black lines mark the 3x3 blocks.

Same grammar, training set, sampler and seed in both arms; only the window rule differs. The criterion starts at k=9 and backs off as the grid fills; the control is pinned at the best fixed k=7 for the whole run.

	success	gram	novel
criterion	0.4062	0.6250	0.7812
fixed k=7	0.0625	0.0938	0.9688

The win is grammaticality (6.7x here), bought with a little novelty: the fixed window is novel almost always because it is mostly emitting grids that no root derives.

Where this leaves the patch-size question

what was tried	effect	support	reading
choose the window adaptively vs any fixed k	12x	12/12 cells	the length scale is the whole story
add hshunt's entropy floor h_min > 0	worse	10/10 cells	h_min=0 (bare support) is best everywhere
add the count floor n_min > 1	worse	both P rungs	same mechanism: rejection shortens the window
replace per-site choice with its own k(t)	15% kept	9/9 cells	at matched mean k -- same budget, worse spent
predict k from position (block offset)	R^2<0.001	18/18 runs	no positional signature at all
predict k from local reveal count	R^2<=0.47	18/18 runs	subsumes time; rest is content-dependent

Answer to "how are you setting the patch size over time?"

> Not over time. The rule that works is "the largest window that matches any training patch at all", evaluated SEPARATELY AT EVERY SITE. Its k(t) curve is a summary of what it does, not a specification of it: handing that same curve back to the sampler recovers only 15% of the success.
>
> The entropy criterion in particular does not bind -- every floor above zero costs success in every cell measured, because a rejected window falls back on a shorter one rather than on a better one.

Caveat: all of this is s=3, L=2 (9x9, rungs 5/7/9), v in {2,3,4}, m=2 mostly, n_gen=4096. Deeper boards need k >= U1 = 2s^(L-1)-1, which is out of reach at L=4.